

Solidissima vs Mercenaria synteny (gene indices) Solidissima vs Mercenaria synteny (chromosome coordinates)



	Solidissima scaf	Mercenaria scaf	p (FET)	Count
0	Chr01 : 0-end	NC_069362.1 : 0-end	0.00e+00	1071
1	Chr03 : 0-end	NC_069361.1 : 0-end	0.00e+00	1059
2	Chr06 : 0-end	NC_069363.1 : 0-end	0.00e+00	941
3	Chr05 : 0-end	NC_069366.1 : 0-end	0.00e+00	822
4	Chr08 : 0-end	NC_069373.1 : 0-end	0.00e+00	808
5	Chr04 : 0-end	NC_069364.1 : 0-end	0.00e+00	795
6	Chr02 : 0-end	NC_069365.1 : 0-end	0.00e+00	766
7	Chr07 : 0-end	NC_069372.1 : 0-end	0.00e+00	746
8	Chr14 : 0-end	NC_069377.1 : 0-end	0.00e+00	672
9	Chr09 : 0-end	NC_069370.1 : 0-end	0.00e+00	672
10	Chr10 : 0-end	NC_069369.1 : 0-end	0.00e+00	665
11	Chr13 : 0-end	NC_069368.1 : 0-end	0.00e+00	644
12	Chr11 : 0-end	NC_069375.1 : 0-end	0.00e+00	530
13	Chr16 : 0-end	NC_069371.1 : 0-end	0.00e+00	521
14	Chr12 : 0-end	NC_069374.1 : 0-end	0.00e+00	517
15	Chr15 : 0-end	NC_069367.1 : 0-end	0.00e+00	506
16	Chr17 : 0-end	NC_069376.1 : 0-end	0.00e+00	443
17	Chr19 : 0-end	NC_069378.1 : 0-end	0.00e+00	300
18	Chr18 : 0-end	NC_069379.1 : 0-end	0.00e+00	210
19	ctg177 : 0-end	NC_048487.1 : 0-end	5.01e-07	3
20	ctg116 : 0-end	NC_048487.1 : 0-end	1.46e-03	2

(a) depicts the chromosomes/scaffolds of *Solidissima* (x-axis) plotted against the chromosomes/scaffolds of *Mercenaria* (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 14533 orthologs found between 27 *Solidissima* scaffolds and 271 *Mercenaria* scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. *Nature Ecology & Evolution* 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.